

Syllabus
Academic Year 2024-2025

1. General Information	
Course title	Introduction to Optimization Theory
Degree cycle (level)/major	Bachelor, Mathematical and Computational Science
Year, term	3, 2
Number of credits	5
Language of delivery:	English
Prerequisites	Calculus1 – Calculus4, Linear Algebra
Postrequisites	Operations Research, Machine learning
Lecturer(s)	Nurseitova Altyn, The Candidate of Physical and Mathematical Sciences, Associate Professor, altyn.nurseitova@astanait.edu.kz , Astana IT University, Expo, C1 block, 3rd floor, office C1.1.336
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	Introduction to Optimization Theory plays a critical role in understanding a wide range of fields, including economics, engineering, data science, machine learning, and operations research. This course covers the fundamental concepts and methods in optimization, including the mathematical formulation of optimization problems, linear and nonlinear optimization techniques, and constrained optimization.
2. Course Goal(s)	After completing this course, students should have developed a clear understanding of the fundamental concepts in optimization theory and a range of skills that allow them to effectively solve optimization problems. The main goal is to be able to apply optimization techniques, including linear programming, gradient methods, and constrained optimization, to problems in post-requisite courses.
3. Course Objectives:	- To introduce students to the basic principles and techniques in optimization theory, including the formulation and classification of optimization problems. - To equip students with the skills necessary to analyze and solve linear and nonlinear optimization problems, both with and without constraints. - To provide students with practical experience in applying optimization algorithms, such as the Simplex method for linear programming, gradient descent for unconstrained optimization, and Lagrange multipliers for constrained optimization. - To familiarize students with convex optimization problems and the associated theory, including duality and optimality conditions, to streamline the problem-solving process.

4. Skills & Competences	• Proficiency in transforming word problems into mathematical optimization problems, both linear and nonlinear. • Mastery of optimization algorithms, including the Simplex method for linear programming, gradient descent for unconstrained optimization, and techniques for solving constrained optimization problems. • Familiarity with solving convex optimization problems and using duality to simplify complex problems. • Ability to critically assess the structure of optimization problems and identify appropriate solution methods based on the problem's nature (e.g., linearity, convexity, and constraints). • Capability to analyze the feasibility and optimality conditions of an optimization problem and interpret the results accurately. • Ability to identify potential issues, such as non-convergence or local optima, and take corrective actions or make informed decisions about solution reliability.
5. Course Learning Outcomes:	By the end of this course students will be able to: • Translate real-world problems into mathematical optimization models, identifying objective functions, constraints, and decision variables in both linear and nonlinear contexts. • Apply the Simplex method and other algorithms to solve linear programming problems and interpret the results effectively. • Use gradient-based methods (e.g., gradient descent) to solve unconstrained optimization problems and understand their convergence properties. • Utilize methods like Lagrange multipliers and the Karush-Kuhn-Tucker (KKT) conditions to solve optimization problems with constraints. • Identify convex problems and apply relevant theoretical results, such as duality, to find optimal solutions efficiently. • Clearly present optimization solutions and their interpretations in both written and oral formats, using appropriate mathematical notation and language.
6. Methods of Assessment	• Homework • Classwork assignments • Quiz • Midterm Exam • Endterm Exam • Final Exam
7. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). <u>Basic Literature:</u> 1. Lecture presentations. 2. <i>An Introduction to Optimization</i>, by Edwin K. P. Chong and Stanislaw H. Zak, A Wiley-Interscience Publication, John Wiley

	& Sons, Inc. New York / Chichester / Weinheim / Brisbane / Singapore / Toronto. (Any edition). <u>Supplementary literature:</u> 1. <i>Introduction to Optimization</i>, by Boris T. Polyak, Optimization Software, Inc., Publications Division, New York, 2020.
1. Resources	Online journals, article, papers, books and internet resources as well as online emulators and online software for simulation.
2. Course policy	Course and University policies include: Attendance: Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as “not graded”. In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean’s office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing. Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student’s misconduct, the student can be dispensed from the classes. Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the

	being late to the class is not welcome and can have restriction activities by the course instructor. Attestation I and II: Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course. Home work / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures. <i>In case the homework/assignments are submitted in handwritten form (for example, a photo of a handwritten copy), all unreadable or illegible work will be considered failed (scored as 0).</i> Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale. <i>In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.</i> Final exam* / Final project: At the completion of this course, the Final Exam will be given in the written form Laptops and mobile devices can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor. Online lessons can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. Cheating and plagiarism are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data;
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	4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.
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** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	PA	Practical assignment
4	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	Fundamentals of Optimization. Basic concepts: optimization, objective functions, constraints. Classification of optimization problems. Mathematical foundations for optimization.	3	2	1	9
2	Linear Programming. Formulation of linear programming problems. Simplex method and its principles.	3	2	1	9
3	Duality in Linear Programming. Primal and dual problems. Strong and weak duality theorems.	3	2	1	9
4	Nonlinear Unconstrained Optimization. Necessary and sufficient conditions for local extrema. Gradient descent and Newton's methods.	3	2	1	9

5	The influence of noise. Sources and types of noise. The gradient method in the presence of noise.	3	1	1	9
5	Midterm Exam		1		
6	Nonlinear Constrained Optimization: Equality Constraints. Problem formulation. Lagrange multiplier condition, and second-order conditions. Minimizing quadratic functions subject to linear constraints.	3	2	1	9
7	Nonlinear Constrained Optimization: Inequality Constraints. Karush-Kuhn-Tucker (KKT) conditions and second-order conditions.	3	2	1	9
8	Convex Optimization. Properties of convex optimization problems. Convex sets, convex functions, and their role in optimization.	3	2	1	9
9	Algorithms for Constrained Optimization. Projections. Projected gradient methods. Penalty methods.	3	2	1	9
10	Derivative-Free Optimization Algorithms. Unconstrained Optimization and Neural Networks. Genetic Algorithms.	3	1	1	9
10	Endterm Exam		1		
	Total hours:	30...	20	10	90

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
1	Study the basic concepts of optimization, objective functions, and constraints. Classify different types of optimization problems (linear, nonlinear, constrained, unconstrained). Review the mathematical foundations required for optimization, including calculus and linear algebra.	9	Reading list, books, internet resources Electronic version	Electronic version or hard copy of the assignment
2	Study the formulation of linear programming problems. Understand the Simplex method and its principles. Solve linear programming problems using the Simplex algorithm.	9	Reading list, books, internet resources Electronic version	Electronic version or hard copy of the assignment

3	Understand the concepts of primal and dual problems in linear programming. Study the Strong and Weak Duality Theorems. Solve example problems using dual formulations.	9	Reading list, books, internet resources Electronic version	Electronic version or hard copy of the assignment
4	Study the necessary and sufficient conditions for local extrema in unconstrained optimization. Implement and compare the Gradient Descent and Newton's methods for simple unconstrained optimization problems.	9	Reading list, books, internet resources Electronic version	Electronic version or hard copy of the assignment
5	Study the different sources and types of noise in optimization (e.g., measurement errors, random fluctuations). Analyze the effect of noise on optimization algorithms like the gradient method.	9	Reading list, books, internet resources Electronic version	Electronic version or hard copy of the assignment
6	Study problem formulation and Lagrange multiplier condition for nonlinear optimization with equality constraints. Derive and apply second-order conditions for optimality.	9	Reading list, books, internet resources Electronic version	Electronic version or hard copy of the assignment
7	Study the Karush-Kuhn-Tucker (KKT) conditions and their role in solving nonlinear constrained optimization problems with inequality constraints. Derive and apply second-order KKT conditions.	9	Reading list, books, internet resources Electronic version	Electronic version or hard copy of the assignment
8	Study the properties of convex optimization problems, including convex sets, convex functions, and their importance in optimization. Solve simple convex optimization problems.	9	Reading list, books, internet resources Electronic version	Electronic version or hard copy of the assignment
9	Study projection-based methods for constrained optimization. Implement the projected gradient method for constrained problems. Learn about penalty methods for handling constraints in optimization.	9	Reading list, books, internet resources Electronic version	Electronic version or hard copy of the assignment
10	Study and implement derivative-free optimization methods such as Genetic Algorithms for unconstrained optimization	9	Reading list, books, internet resources Electronic version	Electronic version or hard copy of the assignment

	problems.			
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4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1 st attestation	Assignments**: Homework 1, Homework 2, Quiz 1, Mid Term	60 20 20 20 40	100
2 nd attestation	Assignments**: Homework 3, Homework 4, Quiz 2, End Term	60 20 20 20 40	100
Final exam*	Written Exam		100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final		100

****** The number of assignments can be different. It depends from the course program and designed by course syllabus. But the total points of the assignment are 60 in each control period.

***** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well.

Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Nurseitova Altyn	Associate Professor	29/11/2024	

Director

Sergaziyev M.Zh.